δ-potential via momentum space S.E.

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx/\hbar} \psi(x)\, dx$$

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{ipx/\hbar} \phi(p)\, dp$$

S.E. $\quad -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} - \alpha\, \delta(x)\, \psi(x) = E\, \psi(x) \qquad$ bound state $E = -|E| < 0$

Multiply by $\dfrac{e^{-ipx/\hbar}}{\sqrt{2\pi\hbar}}$ and integrate over x:

$$\frac{1}{\sqrt{2\pi\hbar}} \left(-\frac{\hbar^2}{2m}\right) \int_{-\infty}^{\infty} e^{-ipx/\hbar} \frac{d^2\psi}{dx^2}\, dx - \frac{\alpha}{\sqrt{2\pi\hbar}} \int \delta(x)\, e^{-ipx/\hbar} \psi(x)\, dx$$

$$= -|E| \int \frac{e^{-ipx/\hbar}}{\sqrt{2\pi\hbar}} \psi(x)\, dx$$

By parts: $\displaystyle\int_{-\infty}^{\infty} e^{-ipx/\hbar} \frac{d^2\psi}{dx^2}\, dx = \cancel{e^{-ipx/\hbar} \frac{d\psi}{dx}\Big|_{-\infty}^{+\infty}} - \int \frac{d e^{-ipx/\hbar}}{dx} \frac{d\psi}{dx}\, dx$

$$= -\cancel{\frac{d e^{-ipx/\hbar}}{dx} \psi \Big|_{-\infty}^{+\infty}} + \int_{-\infty}^{\infty} \frac{d^2 e^{-ipx/\hbar}}{dx^2} \psi(x)\, dx =$$

$$= \int \left(\frac{-ip}{\hbar}\right)^2 e^{-ipx/\hbar} \psi(x)\, dx$$

$\Rightarrow$ S.E. $\quad -\frac{\hbar^2}{2m} \frac{-p^2}{\hbar^2} \phi(p) - \alpha \underbrace{\int \delta(x) e^{-ipx/\hbar} \psi(x) \frac{dx}{\sqrt{2\pi\hbar}}}_{\frac{\psi(0)}{\sqrt{2\pi\hbar}} = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \phi(p)\, dp = \frac{C}{2\pi\hbar}} = -|E|\, \phi(p)$

SE $\quad \frac{p^2}{2m} \phi(p) - \frac{\alpha}{2\pi\hbar} C = -|E|\, \phi(p)$

$$\Rightarrow \phi(p) = \frac{\frac{\alpha}{2\pi\hbar} C}{\frac{p^2}{2m} + |E|} = \frac{m\alpha}{\pi\hbar} \frac{C}{p^2 + 2m|E|}$$

But $\quad C = \int_{-\infty}^{\infty} \phi(p)\, dp = \frac{m\alpha}{\pi\hbar} C \int_{-\infty}^{\infty} \frac{dp}{p^2 + 2m|E|}$

Divide by C.

$$\Rightarrow \quad 1 = \frac{m\alpha}{\pi\hbar} \int_{-\infty}^{\infty} \frac{dp}{p^2 + 2m|E|} = \frac{m\alpha}{\pi\hbar} \frac{\pi}{\sqrt{2m|E|}}$$

$$\int_{-\infty}^{\infty} \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right)\Big|_{-\infty}^{\infty} = \frac{\pi}{a}$$

$$\Rightarrow \quad 1 = \frac{\alpha}{\hbar}\sqrt{\frac{m}{2}} \frac{1}{\sqrt{|E|}}$$

Solution exists only for $\underline{\underline{\alpha > 0}}$

$$|E| = \frac{\alpha^2}{\hbar^2} \cdot \frac{m}{2} = \frac{m\alpha^2}{2\hbar^2}$$

$$E = -|E| = -\frac{m\alpha^2}{2\hbar^2}$$

the bound state! (same as before)

Find C: $\quad \int |\phi(p)|^2 dp = 1$

$$1 = \left(\frac{m\alpha C}{\pi\hbar}\right)^2 \int_{-\infty}^{\infty} \frac{dp}{\left(p^2 + 2m\frac{m\alpha^2}{2\hbar^2}\right)^2} = \left(\frac{m\alpha C}{\pi\hbar}\right)^2 \int_{-\infty}^{\infty} \frac{dp}{\left(p^2 + \frac{m^2\alpha^2}{\hbar^2}\right)^2}$$

Note that we need the integral

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^2} = -\frac{d}{d(a^2)} \int_{-\infty}^{\infty} \frac{dx}{x^2 + a^2} = -\frac{d}{d(a^2)} \frac{\pi}{\sqrt{a^2}} =$$

$$= (-\pi)\left(-\frac{1}{2}\right)(a^2)^{-3/2} = \frac{\pi}{2a^3}$$

$$\Rightarrow \quad 1 = \left(\frac{m\alpha C}{\pi\hbar}\right)^2 \frac{\pi}{2\left(\frac{m\alpha}{\hbar}\right)^3} = \frac{C^2}{\pi^2} \frac{\pi}{2} \frac{\hbar}{m\alpha}$$

$$\Rightarrow \quad c^2 = \frac{2\pi m\alpha}{\hbar}, \quad c = \sqrt{\frac{2\pi m\alpha}{\hbar}}$$

$$\phi(p) = \sqrt{\frac{2}{\pi}\left(\frac{m\alpha}{\hbar}\right)^3}\ \frac{1}{p^2 + \frac{m^2\alpha^2}{\hbar^2}}$$

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int \phi(p)\, e^{ipx/\hbar}\, dp = \qquad \left(p = \frac{m\alpha}{\hbar} q\right)$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \left(\frac{m\alpha}{\hbar}\right) \sqrt{\frac{2}{\pi}\left(\frac{m\alpha}{\hbar}\right)^3}\ \frac{1}{\left(\frac{m\alpha}{\hbar}\right)^2} \int_{-\infty}^{\infty} \frac{e^{i\frac{m\alpha}{\hbar^2} q x}\, dq}{q^2+1}$$

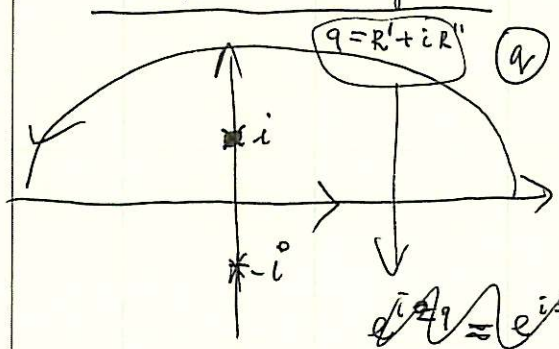
$$= \frac{1}{\sqrt{2\pi\hbar}} \left[\left(\frac{\hbar}{m\alpha}\right)^2 \frac{2}{\pi}\left(\frac{m\alpha}{\hbar}\right)^3\right]^{1/2} \int_{-\infty}^{\infty} \frac{e^{iqz}\, dq}{q^2+1} \qquad z = \frac{m\alpha}{\hbar^2} x$$

Now need $I(z) = \displaystyle\int_{-\infty}^{\infty} \frac{\cos qz + i \sin qz}{q^2+1}\, dz =$

$$= \int_{-\infty}^{\infty} \frac{\cos qz}{q^2+1}\, dz = \mathrm{Re} \int_{-\infty}^{\infty} \frac{e^{iqz}\, dq}{q^2+1}$$

Clearly $I(z) = I(|z|)$ as $\cos z$ = even.

<u>Contour integration</u>



$\cancel{I(z) = 2\pi i \sum \mathrm{Res}}$

<u>Cauchi Theorem</u>

$\cancel{e^{izq} = e^{izR' - zR''} \to 0 \quad R'' \to \infty}$

$$I(z) = 2\pi i \sum_s \mathrm{Res}(\ )$$

$q = Re^{i\varphi}$

$$\int \frac{e^{iqRe^{i\varphi}}\, R\, de^{i\varphi}}{R^2 e^{i2\varphi}} = \frac{1}{R} \xrightarrow[R\to\infty]{} 0$$

$$\text{Res} = \text{Res}\,\frac{e^{iqz}}{(q+i)(q-i)} = \frac{e^{iz(i)}}{2i}$$

$$I(z) = 2\pi i\,\frac{e^{-z}}{2i} = \pi e^{-z}$$

$$\Rightarrow \psi(x) = \frac{1}{\sqrt{2\pi\hbar}}\sqrt{\frac{2}{\pi}\,\frac{m\alpha}{\hbar}}\;\pi\, e^{-\frac{m\alpha}{\hbar^2}|x|} = \sqrt{\frac{m\alpha}{\hbar^2}}\; e^{-\frac{m\alpha}{\hbar^2}|x|}$$

exactly as in Eq. (2.129) !